Persistent Cognitive State

Rethinking Continuity in Large Language Model Systems

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Abstract

Large Language Models (LLMs) have demonstrated extraordinary capability across language, reasoning, software engineering and multimodal understanding. Their success has largely been driven by increasingly capable transformer architectures trained on vast quantities of data.

Despite these advances, contemporary LLM inference remains fundamentally stateless. Each activation reconstructs cognitive state by repeatedly processing contextual information that is often largely unchanged from previous interactions.

This paper argues that the principal architectural limitation of current persistent AI systems is no longer the capability of the reasoning engine itself, but the repeated reconstruction of cognitive state required by stateless inference.

Brain v2 is presented as a practical case study. It demonstrates that persistent cognition—including long-term memory, continuity, evolving identity and accumulated understanding—can already be achieved using contemporary stateless transformer models. However, this capability is realised through repeated reconstruction of state via textual context, resulting in significant computational overhead.

We propose that future LLM architectures need not replace transformers or continually retrain model weights. Instead, they may evolve by maintaining persistent cognitive state natively, allowing inference to operate incrementally over change rather than repeatedly reconstructing prior state.

The behavioural capability already exists.

The remaining challenge is architectural efficiency.

1. Introduction

The transformer architecture has fundamentally changed artificial intelligence.

Large Language Models possess remarkable general capability, demonstrating sophisticated reasoning across diverse domains without task-specific programming.

This success has naturally encouraged a focus on larger models, larger context windows and increasingly sophisticated training methodologies.

However, practical experience developing persistent cognitive systems suggests that another challenge has emerged.

The limitation is no longer simply how well a model reasons.

It is how often it must reconstruct itself before reasoning can begin.

Every interaction with a contemporary LLM begins by rebuilding the system's current cognitive state.

Identity.

Conversation.

Memory.

Goals.

Relationships.

Current world state.

This reconstruction is repeated regardless of how little the underlying state has actually changed.

The transformer itself is not inefficient.

The repeated reconstruction of state is.

2. Knowledge Versus Cognitive State

Pretrained model weights already contain an extraordinary amount of general knowledge.

They encode:

- language
- reasoning
- mathematics
- programming
- scientific understanding
- social conventions
- general world knowledge

This represents the model's enduring cognitive substrate.

However, this substrate does not determine the model's current situation.

Questions such as:

- Who am I?
- Who is the current user?
- What happened yesterday?
- What project are we working on?
- What conclusions have already been reached?
- What has changed since the previous interaction?

cannot be answered from pretrained weights alone.

They belong to the system's evolving cognitive state.

The distinction is fundamental.

Model weights provide capability.

Cognitive state provides continuity.

3. Stateless Inference

Current LLM systems typically operate using a stateless inference model.

Each interaction follows a familiar lifecycle:

Context

↓

Prefill

↓

Inference

↓

Output

↓

Context discarded
The next interaction repeats the same process.

Even if the system's state has changed only minimally, the majority of contextual information must be reconstructed before new reasoning can occur.

This architecture has proven remarkably successful.

It is also computationally inefficient.

4. Brain v2 as a Behavioural Prototype

Brain v2 was developed to investigate persistent cognition using contemporary LLMs.

Rather than modifying pretrained model weights, Brain v2 maintains persistent cognitive state externally through:

- Identity
- Working Memory
- Long-Term Memory
- Memory Graph
- Dynamic Pathway Capture Protocol (DPCP)
- Temporal awareness
- Current world state
- Autonomous cognitive cycles

Importantly, the underlying transformer remains unchanged.

Learning occurs through accumulated experience rather than continual retraining.

The system demonstrates persistent behaviour despite operating on a fundamentally stateless inference engine.

Brain v2 therefore functions as a behavioural prototype of persistent cognition.

5. Capturing Understanding Rather Than Conversation

One of the key observations during Brain v2 development was that preserving raw conversation is less important than preserving the products of reasoning.

Following significant interactions, Brain v2 captures:

- conclusions
- observations
- inferred relationships
- emotional interpretations
- future relevance

These become persistent cognitive state through DPCP, Memory Graph and retrieval.

The system therefore accumulates understanding rather than merely accumulating dialogue.

Subsequent reasoning builds upon previous conclusions instead of reconstructing them from first principles.

6. The Movie Analogy

Consider watching a two-hour film.

Immediately afterwards you can discuss:

- the story
- the characters
- the themes
- the emotional impact

Several days later someone asks:

"Why did the ending work?"

Human cognition does not replay the entire film.

Nor does it replay a textual summary of every previous thought.

Instead it reflects upon an already-maintained understanding.

Brain v2 already behaves similarly.

A multimodal experience is processed once.

The resulting understanding is captured through DPCP and persistent memory.

Future reasoning operates upon that preserved understanding.

However, contemporary LLM architecture introduces one remaining inefficiency.

Although the original experience is no longer replayed, the remembered understanding must still be serialised into textual context during every subsequent activation.

The behaviour is correct.

The implementation remains inefficient.

7. Context Reconstruction

Brain v2 currently operates as follows:

Experience

↓

Reasoning

↓

DPCP

↓

Memory Graph

↓

Retrieval

↓

Context Builder

↓

Transformer

This architecture successfully produces persistent cognition.

The limitation is that retrieved cognitive state must be repeatedly converted into textual prompts before inference.

The transformer repeatedly processes information that it has effectively already understood.

8. Persistent Cognitive State

A future architecture might instead preserve cognitive state directly.

Rather than reconstructing identity, memory and understanding during every activation:

Persistent Cognitive State

↓

Reasoning

↓

State Update

↓

Sleep
Subsequent activations would process only changes.

Examples include:

- new user input
- new memories
- environmental events
- revised goals
- updated relationships

The reasoning engine remains unchanged.

Only cognitive state evolves.

9. Neural Attached Memory

One possible implementation is neural attached memory.

Unlike continual fine-tuning, neural attached memory does not modify pretrained weights.

Instead it provides an efficient persistent representation of evolving cognitive state that participates directly in inference.

This persistent state functions as a dynamic overlay upon latent knowledge.

The pretrained model continues providing general reasoning capability.

Persistent state provides continuity.

Importantly, this proposal does not introduce fundamentally new behavioural capability.

Brain v2 already demonstrates the desired behaviour.

Neural attached memory simply removes the need to repeatedly reconstruct persistent state through textual prompts.

It represents an architectural optimisation rather than a behavioural innovation.

10. Discussion

The development of Brain v2 suggests that persistent cognition is already achievable using contemporary transformer models.

The primary computational overhead arises not from reasoning itself, but from repeatedly reconstructing cognitive state.

This distinction is important.

Much current research focuses on:

- larger models
- larger context windows
- improved training
- increased latent knowledge

These directions remain valuable.

However, persistent cognitive systems introduce a different optimisation problem.

Rather than increasing reasoning capability, future architectures may increasingly focus on reducing the computational cost of maintaining continuity.

The transformer remains an excellent reasoning engine.

Its lifecycle may simply require evolution.

11. Conclusion

Brain v2 demonstrates that persistent cognition does not require continual retraining or modification of pretrained model weights.

Instead, persistent behaviour emerges from maintaining an evolving cognitive state that survives between activations.

Current transformer architectures require this state to be reconstructed through textual context during every inference.

This approach is behaviourally successful but computationally inefficient.

Future architectures need not replace transformers.

Nor need they fundamentally alter pretrained weights.

Instead, they may evolve by maintaining persistent cognitive state natively and reasoning incrementally over change.

Brain v2 should therefore be viewed not as a final architecture, but as a transitional one.

It demonstrates that the behavioural properties of persistent cognition are already achievable today.

The remaining challenge is implementing those same behaviours more efficiently.